

One Countable Non-Dense Scalar Set for Every Nonexceptional Supercyclic Spectral Class

Candidate sharp spectral theorem for arXiv:1509.04912

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Status. Candidate full theorem, likely valid; human review recommended. It gives a sharp maximal spectral partial answer to Question 3: one explicit countable non-dense scalar set works in every supercyclic adjoint-spectral class except $\{1\}$, and no non-dense set can cover that last class uniformly. It does not characterize every admissible scalar set.

1 Source problem

The source is S. Charpentier, R. Ernst, and Q. Menet, Γ -supercyclicity, arXiv:1509.04912 [1]. For $T \in \mathcal{L}(X)$, a vector x is Γ -supercyclic when

$$\text{Orb}(\Gamma x, T) = \{\gamma T^n x : \gamma \in \Gamma, n \geq 0\}$$

is dense in X . Question 3 asks which $\Gamma \subset \mathbb{C}$ make supercyclicity equivalent to Γ -supercyclicity.

Theorem (Shkarin Theorem). *Let X be a complex Banach space and $T \in L(X)$. T is $\mathbb{R}^+$ -supercyclic if and only if T is supercyclic and either the point spectrum $\sigma_p(T^*)$ is empty or $\sigma_p(T^*) = \{re^{i\theta}\}$, with $r \neq 0$ and $\theta \in \pi(\mathbb{R} \setminus \mathbb{Q})$.*

More generally the following question arises, an answer to which would involve the spectrum of T^* .

Question 3. For which $\Gamma \subset \mathbb{C}$ is supercyclicity equivalent to Γ -supercyclicity?

1.2. Statements of the main results. The purpose of this article is to discuss Questions 1, 2 and 3. In order to deal with Questions 1 and 2 we introduce two properties that a subset of $\mathbb{C}$ can enjoy or not.

Figure 1: Question 3 on page 3 of the source paper.

The source exactly treats each unimodular adjoint eigenvalue in Theorem C, but explicitly leaves the nonunit and empty-adjoint-point-spectrum regimes unclear. The theorem below fills all of those regimes at once with the same small scalar set and simultaneously handles every unimodular eigenvalue except the uniquely unavoidable value 1.

Furthermore we give a general answer to Question 3 when $\sigma_p(T^*)$ is non-empty.

Theorem C. *Let X be a complex Banach space.*

- (1) *For every $\theta \in \mathbb{R}$, $\Gamma \subset \mathbb{C}$ satisfies the property: "For every $T \in L(X)$ with $\sigma_p(T^*) = \{e^{i\theta}\}$ and every $x \in X$*

x is Γ -supercyclic for T if and only if x is supercyclic for T "

if and only if ΓG_θ is dense in $\mathbb{C}$, where G_θ stands for the subgroup of $\mathbb{T}$ generated by $e^{i\theta}$.

- (2) *For any $r \neq 1$ and any $\theta \in \mathbb{R}$, there exist $T \in L(X)$ with $\sigma_p(T^*) = \{re^{i\theta}\}$ and $\Gamma \subset \mathbb{C}$ satisfying $\overline{\Gamma G_\theta} = \mathbb{C}$, such that T is supercyclic but not Γ -supercyclic.*

Remark 1.3. (1) For every $\theta \in \mathbb{R}$, Theorem C provides a complete understanding of the problem of describing those subsets Γ of $\mathbb{C}$ such that for every X , every $T \in L(X)$ with $\sigma_p(T^*) = \{e^{i\theta}\}$ and every $x \in X$,

x is Γ -supercyclic for T if and only if x is supercyclic for T .

Nevertheless such a problem for every T (namely for those T such that $\sigma_p(T^*) = \emptyset$ or $\sigma_p(T^*) = \{re^{i\theta}\}$ with $r \neq 1$) remains unclear.

(2) One shall notice that, when $\theta \in \pi(\mathbb{R} \setminus \mathbb{Q})$, the condition $\overline{\Gamma G_\theta} = \mathbb{C}$ is equivalent to $\overline{\Gamma \mathbb{T}} = \mathbb{C}$.

(3) It is worth observing that the equivalence between Γ -supercyclicity and supercyclicity for T with $\sigma_p(T^*) = re^{i\theta}$, $\theta \in \pi(\mathbb{R} \setminus \mathbb{Q})$, depends on r . This differs from the particular case of the $\mathbb{R}_+$ -supercyclicity which is treated in Shkarin Theorem.

Figure 2: Theorem C and the unresolved regimes on page 4 of the source.

2 Statement and proof intuition

Write $\mathbb{Q}(i) = \mathbb{Q} + i\mathbb{Q}$, and set

$$\Gamma_* = \mathbb{Q}(i) \cap (\{z : |z| < 1\} \cup \{z : |z| > 2\} \cup \{z : \Re z > 0\}). \quad (1)$$

Proof intuition

The three pieces of (1) match the three possible adjoint-spectral regimes of a supercyclic operator. When the point spectrum is empty, positive and ordinary supercyclic vectors coincide; positive rationals already suffice and lie in Γ_* . When the eigenvalue is nonunit, the backward-scaled target $\mu^{-n}c$ eventually lies in the inner disk or the exterior region, where Gaussian rationals approximate it with arbitrarily fine relative accuracy. When the eigenvalue is unimodular and not 1, the cyclic rotations of a half-plane cover the whole plane, exactly matching the source's phase-group criterion.

The omitted value 1 is not a weakness of the construction. Theorem C of the source says that a scalar set working uniformly in that class must itself be dense. Hence $\{1\}$ is the unique spectral class that every non-dense uniform construction must exclude.

Theorem 1 (Maximal non-dense spectral coverage). *Let X be an infinite-dimensional complex Banach space. For every $T \in \mathcal{L}(X)$ satisfying $\sigma_p(T^*) \neq \{1\}$ and every $x \in X$,*

$$x \text{ is supercyclic for } T \iff x \text{ is } \Gamma_*\text{-supercyclic for } T.$$

The set Γ_ is countable and non-dense in $\mathbb{C}$.*

Moreover, the exclusion is sharp: if $\Lambda \subset \mathbb{C}$ has the displayed equivalence uniformly for all operators in the class $\sigma_p(T^) = \{1\}$, then Λ is dense in $\mathbb{C}$.*

The adjoint convention in [1] conjugates eigenfunctional multipliers. This causes no ambiguity here: modulus is preserved, 1 is fixed, and a unimodular scalar and its conjugate generate the same cyclic subgroup.

3 A common tail-universal set for every nonunit multiplier

Definition 2. For $\mu \in \mathbb{C} \setminus \{0\}$, a set $\Gamma \subset \mathbb{C}$ is *tail-universal* for μ if

$$\forall c \in \mathbb{C}, \quad \text{dist}(c, \mu^n \Gamma) \longrightarrow 0. \quad (2)$$

Lemma 3 (Universal annular-gap set). *The set Γ_* is tail-universal for every $\mu \neq 0$ with $|\mu| \neq 1$.*

Proof. Fix $c \in \mathbb{C}$. If $c = 0$, use $0 \in \Gamma_*$. Suppose $c \neq 0$ and put $z_n = \mu^{-n}c$. If $|\mu| > 1$, then z_n eventually belongs to the open unit disk. By density of $\mathbb{Q}(i)$, choose $\gamma_n \in \mathbb{Q}(i)$ in that disk with

$$|\gamma_n - z_n| < \frac{|\mu|^{-n}}{n}.$$

Then $\gamma_n \in \Gamma_*$ and $|\mu^n \gamma_n - c| < 1/n$. If $|\mu| < 1$, then $|z_n| > 2$ eventually; choose the same approximation inside the open region $\{|z| > 2\}$. The identical estimate proves (2). $\square$

Lemma 4 (Projective tails). *If x is supercyclic for T , then $\bigcup_{n \geq N} \mathbb{C}T^n x$ is dense in X for every $N \geq 0$.*

Proof. Otherwise some nonempty open set would miss one tail. Density of the full projective orbit would then force a finite union of one-dimensional closed subspaces to be dense in that open set. Being closed, that finite union would contain the open set, contradicting that a finite union of proper linear subspaces has empty interior. $\square$

Proposition 5 (Tail transfer). *Suppose $f \circ T = \mu f$ for some $f \in X^* \setminus \{0\}$ and $\mu \neq 0$. If Γ is tail-universal for μ , then every supercyclic vector for T is Γ -supercyclic.*

Proof. Let x be supercyclic. Necessarily $f(x) \neq 0$. Fix y with $f(y) \neq 0$. Lemma 4 gives $n_k \geq k$ and $\alpha_k \in \mathbb{C}$ such that $\alpha_k T^{n_k} x \rightarrow y$. Applying f yields

$$\alpha_k \mu^{n_k} \rightarrow c := \frac{f(y)}{f(x)} \neq 0.$$

Tail universality supplies $\gamma_k \in \Gamma$ with $\gamma_k \mu^{n_k} \rightarrow c$. Hence

$$\frac{\gamma_k}{\alpha_k} = \frac{\gamma_k \mu^{n_k}}{\alpha_k \mu^{n_k}} \rightarrow 1,$$

so $\gamma_k T^{n_k} x \rightarrow y$. Thus the closure of the Γ -orbit contains $X \setminus \ker f$, a dense set. $\square$

4 Unimodular and empty-spectrum regimes

Lemma 6 (Rotated half-planes). *If $\lambda \in \mathbb{T} \setminus \{1\}$ and G_λ is the cyclic subgroup generated by λ , then $\overline{\Gamma_* G_\lambda} = \mathbb{C}$.*

Proof. The set Γ_* is dense in the open right half-plane, so its closure contains the closed right half-plane $H = \{z : \Re z \geq 0\}$. If λ has finite order $q \geq 2$, the q directions in G_λ have angular gaps $2\pi/q$; every direction is within $\pi/q \leq \pi/2$ of one of them. Thus $\bigcup_{g \in G_\lambda} gH = \mathbb{C}$. If λ has infinite order, G_λ is dense in $\mathbb{T}$ and the same conclusion is immediate. Therefore the closure of $\Gamma_* G_\lambda$ is all of $\mathbb{C}$. $\square$

Lemma 7 (Positive rational reduction). *If x is $\mathbb{R}_+$ -supercyclic for T , then x is $\mathbb{Q}_+$ -supercyclic, and hence Γ_* -supercyclic.*

Proof. The positive projective orbit is dense. Whenever $tT^n x$ lies in a chosen open set, continuity of $s \mapsto sT^n x$ and density of $\mathbb{Q}_+$ in $\mathbb{R}_+$ allow t to be replaced by a positive rational. Finally $\mathbb{Q}_+ \subset \Gamma_*$. $\square$

5 Proof of the main theorem

Proof of Theorem 1. Countability of Γ_* is immediate. Moreover

$$\overline{\Gamma_*} \subset \{|z| \leq 1\} \cup \{|z| \geq 2\} \cup \{\Re z \geq 0\},$$

so it misses the nonempty open half-annulus $\{1 < |z| < 2, \Re z < 0\}$ and is not dense.

Since $\Gamma_* \subset \mathbb{C}$, every Γ_* -supercyclic vector is supercyclic. Conversely, let x be supercyclic for T . The classical adjoint-spectrum dichotomy says that $\sigma_p(T^*)$ is empty or is a singleton $\{\eta\}$ with $\eta \neq 0$ [2].

If it is empty, Proposition LM of [2] (the León-Saavedra–Müller theorem) says that the supercyclic and $\mathbb{R}_+$ -supercyclic vectors coincide. Lemma 7 applies.

Suppose $\sigma_p(T^*) = \{\eta\}$. If $|\eta| \neq 1$, the corresponding eigenfunctional multiplier has nonunit modulus, so Lemma 3 and Proposition 5 apply. If $|\eta| = 1$, the hypothesis gives $\eta \neq 1$; Lemma 6 and the exact unimodular characterization in Theorem C(1) of [1] give that x is Γ_* -supercyclic.

For sharpness, apply the necessity direction of Theorem C(1) in [1] at $\eta = 1$. The generated phase group is $\{1\}$, so uniform equivalence in this spectral class holds only if $\overline{\Lambda} = \mathbb{C}$. $\square$

6 Verification, novelty, and scope

The proof is analytic. The accompanying script checks representative nonunit rescalings and finite cyclic half-plane covers; it is not used as proof.

A bounded search on 9 August 2026 covered:

- the run indexes and arXiv:1509.04912, 1209.1222, 1711.10932, and 2411.03179;
- arXiv queries combining Γ -supercyclicity, countable non-dense scalar set, point spectrum, annulus, and half-plane.

No statement using one countable non-dense set for every spectral class except the sharp eigenvalue-one obstruction was found. Novelty confidence is moderate.

This is a complete and sharp maximal spectral-coverage theorem. It remains a partial answer to Question 3 because it does not characterize all scalar sets for each nonunit eigenvalue or in the empty-spectrum regime. Human review should focus on the three cited spectral inputs, the common tail-universality argument, and the novelty assessment.

References

- [1] S. Charpentier, R. Ernst, and Q. Menet, Γ -supercyclicity, arXiv:1509.04912 (2015).
- [2] S. Shkarin, *Universal elements for non-linear operators and their applications*, J. Math. Anal. Appl. 348 (2008), 193–210; arXiv:1209.1222.